Telecom Customer Churn Prediction using Machine Learning

Abstract

Customer churn prediction is a significant business problem in the telecommunications industry. This project presents a production-grade machine learning system for predicting customer churn using structured telecom data. The system incorporates a complete pipeline including data cleaning, feature preprocessing, exploratory data analysis, class imbalance handling, model training, evaluation, and deployment through an interactive web dashboard. Multiple classification models were implemented and evaluated to provide accurate and interpretable churn predictions.

1 Introduction

Customer churn refers to customers discontinuing services from a company. Telecom providers face substantial revenue losses due to churn, making predictive analytics essential for retention strategies. Machine learning models can detect churn patterns using customer demographics, service subscriptions, and billing characteristics.

The goal of this project is to design a modular churn prediction system capable of accurate classification, visualization, and real-time inference.

2 Dataset Description

The dataset consists of telecom customer records including demographic attributes, subscription details, billing information, and churn labels.

Important numeric features:

- Tenure
- Monthly Charges
- Total Charges

Categorical features include gender, partner status, internet service type, contract type, and payment method.

The target variable Churn is mapped to:

- 1 – Churn
- 0 – Retained

3 Data Loading and Cleaning

A dedicated data ingestion module was developed.

Cleaning steps include:

- Conversion of TotalCharges to numeric
- Missing value imputation using mean
- Removal of customerID identifier
- Target variable encoding

4 Exploratory Data Analysis

A comprehensive EDA module was implemented to analyze dataset characteristics.

The analysis includes:

- Missing value detection
- Class imbalance analysis
- Correlation heatmaps
- Skewness measurement
- Outlier detection using IQR
- Multicollinearity assessment using VIF

5 Feature Preprocessing

A unified preprocessing pipeline was constructed using ColumnTransformer.

5.1 Numeric Processing

- StandardScaler applied to normalize numeric features

5.2 Categorical Processing

- OneHotEncoder applied with unknown category handling

6 Model Development

6.1 Logistic Regression

A Logistic Regression model with lbfgs solver and increased iterations was used to provide interpretable baseline performance.

6.2 Decision Tree

A Decision Tree classifier with entropy criterion was implemented to capture non-linear feature relationships and provide rule-based interpretability.

7 Results and Performance Analysis

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	79.2%	59.4%	68.5%	63.6%
Decision Tree	74.7%	51.6%	76.4%	61.6%

The Logistic Regression model achieved higher accuracy while Decision Tree achieved higher recall, indicating stronger churn detection capability.

8 Model Visual Analysis

8.1 ROC Curve

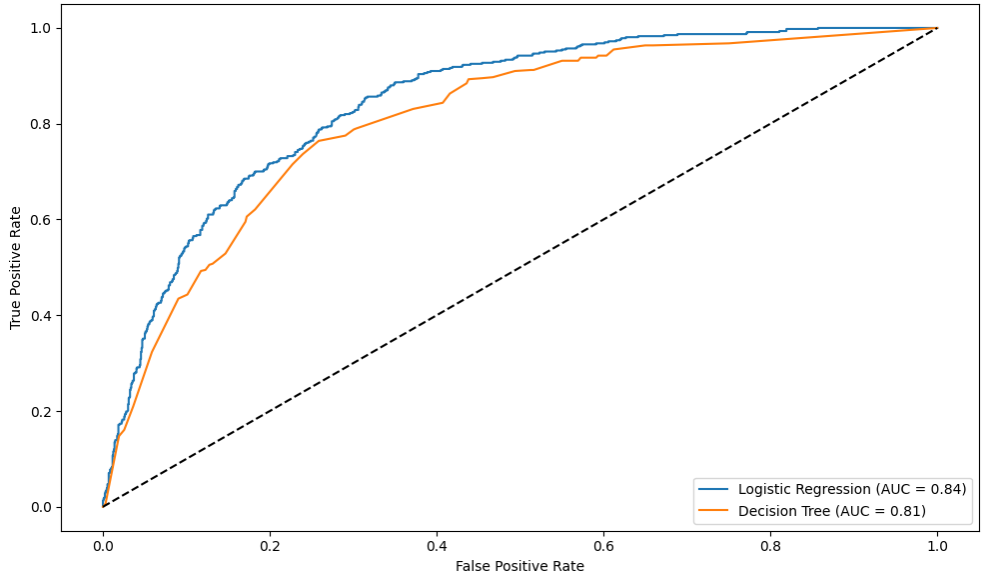

Figure 1: ROC Curve Comparison

8.2 Precision-Recall Curve

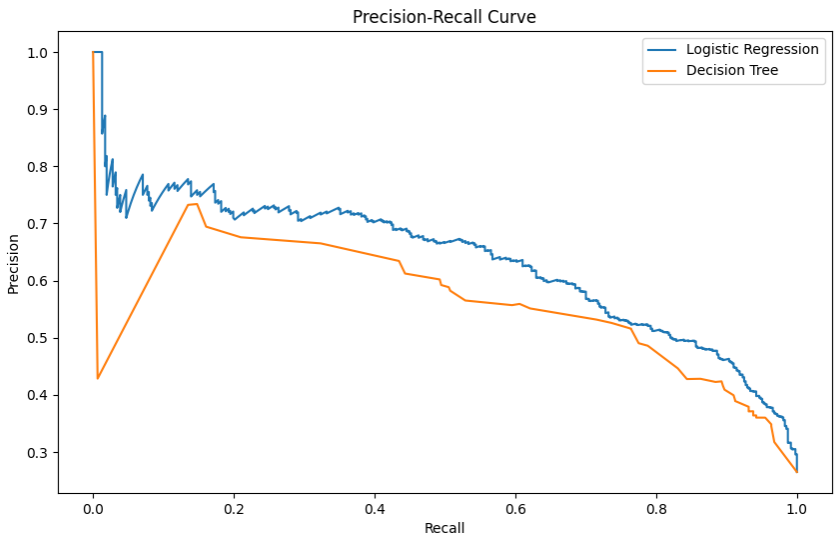

Figure 2: Precision Recall Curve

8.3 Logistic Regression Confusion Matrix

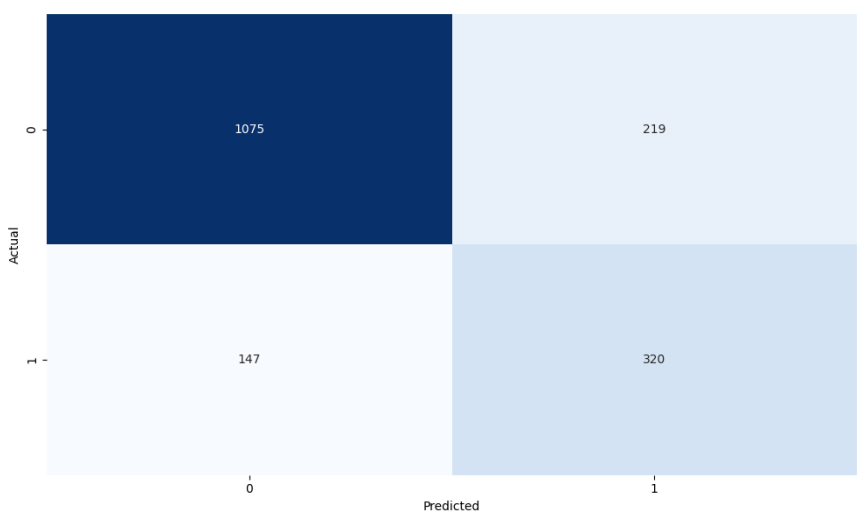

Figure 3: Logistic Regression Confusion Matrix

8.4 Top Logistic Regression Coefficients

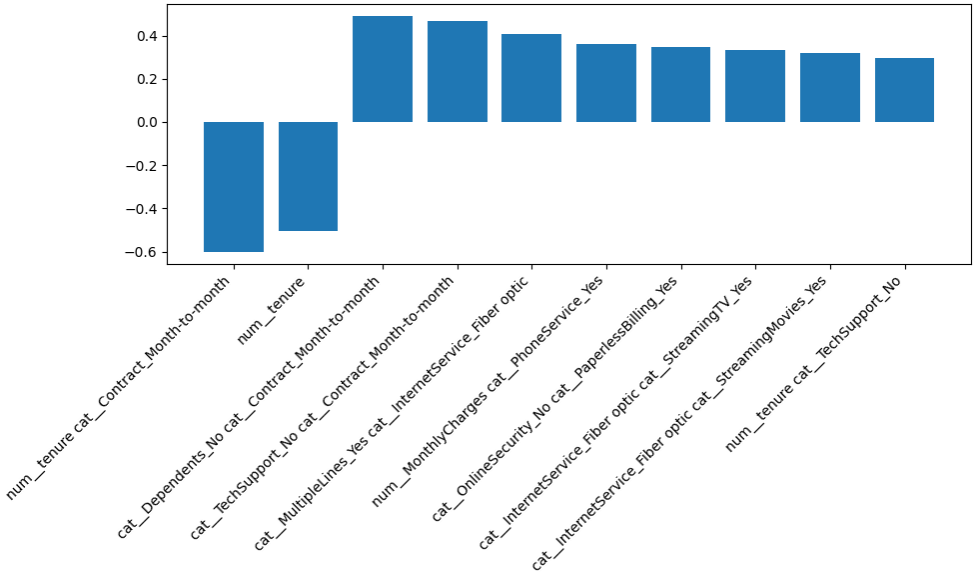

Figure 4: Top Logistic Regression Coefficients

8.5 Decision Tree Confusion Matrix

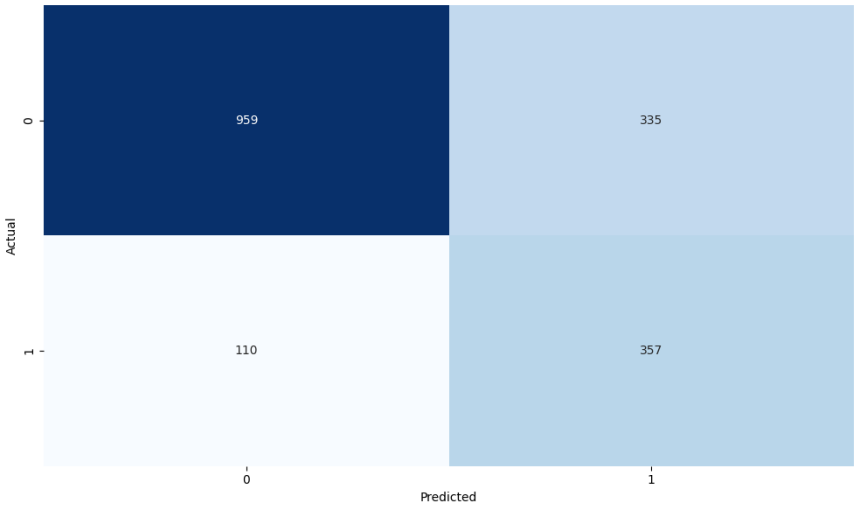

Figure 5: Decision Tree Confusion Matrix

8.6 Decision Tree Feature Importance

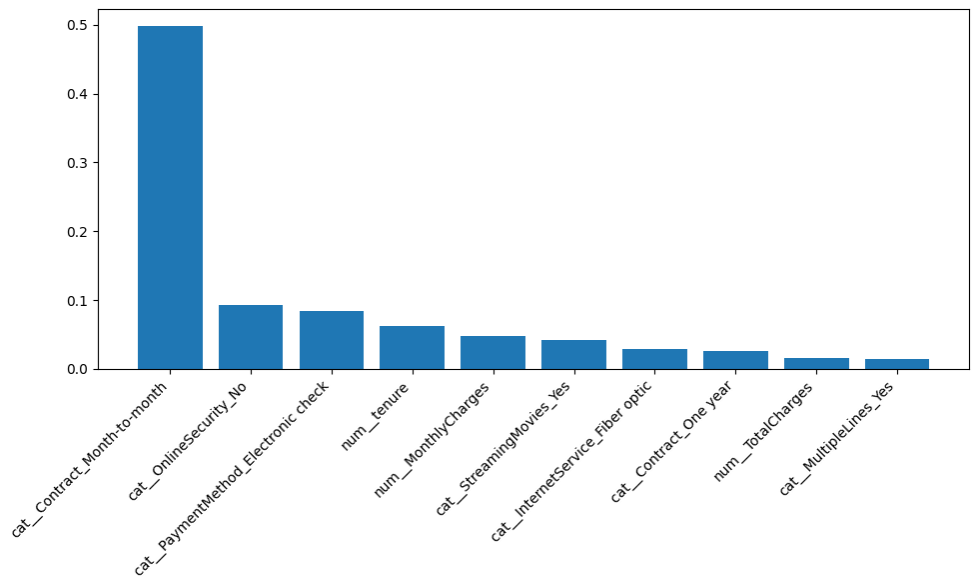

Figure 6: Decision Tree Feature Importance

9 Exploratory Data Analysis Visualizations

9.1 Tenure Distribution

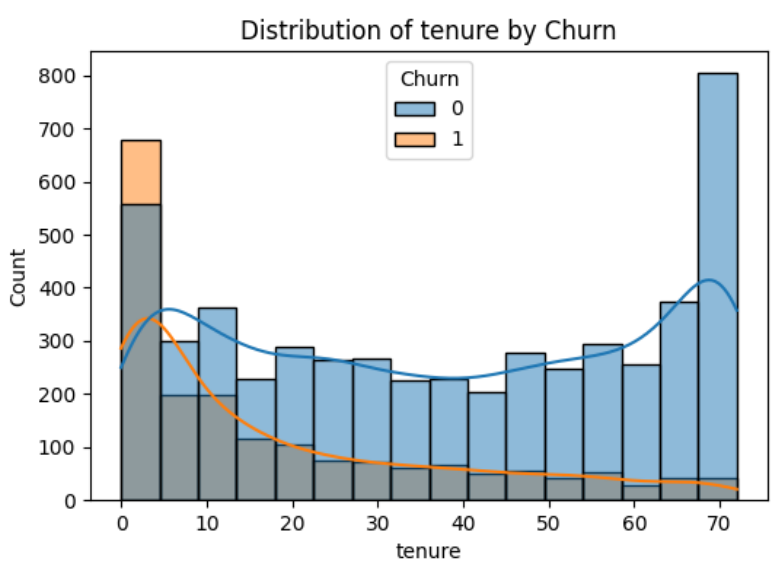

Figure 7: Tenure Distribution

9.2 Monthly Charges Distribution

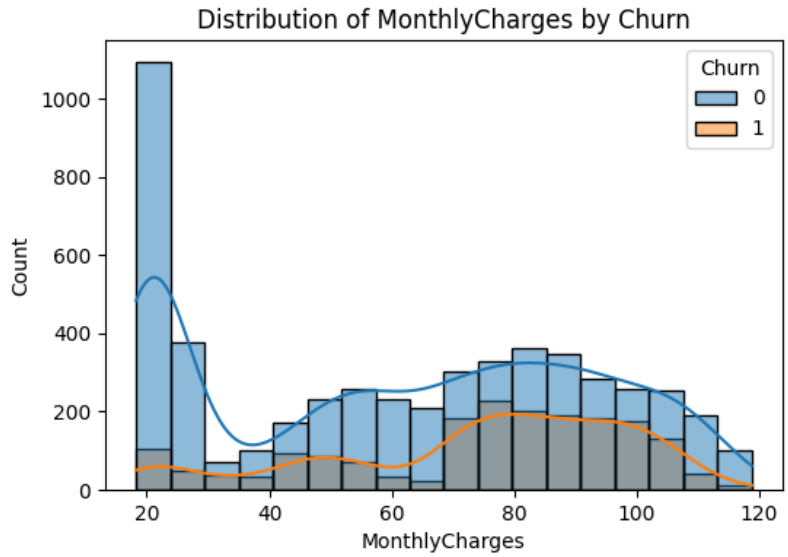

Figure 8: Monthly Charges Distribution

9.3 Total Charges Distribution

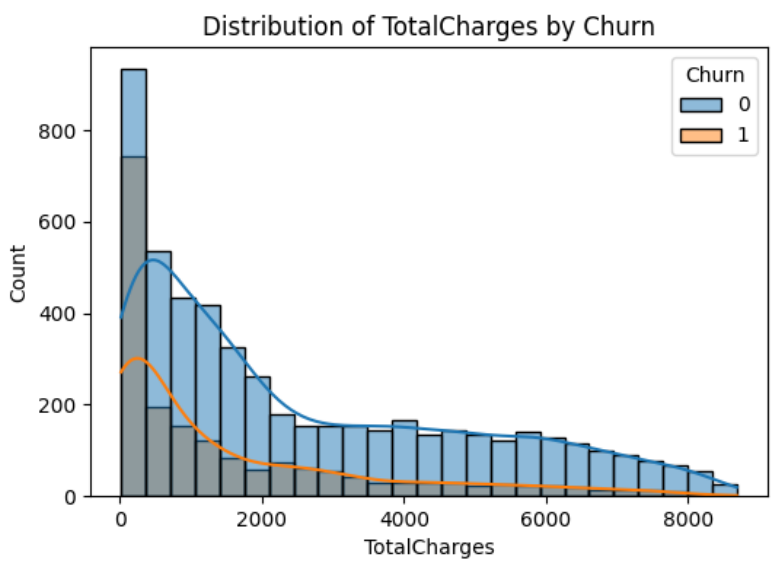

Figure 9: Total Charges Distribution

9.4 Correlation Matrix

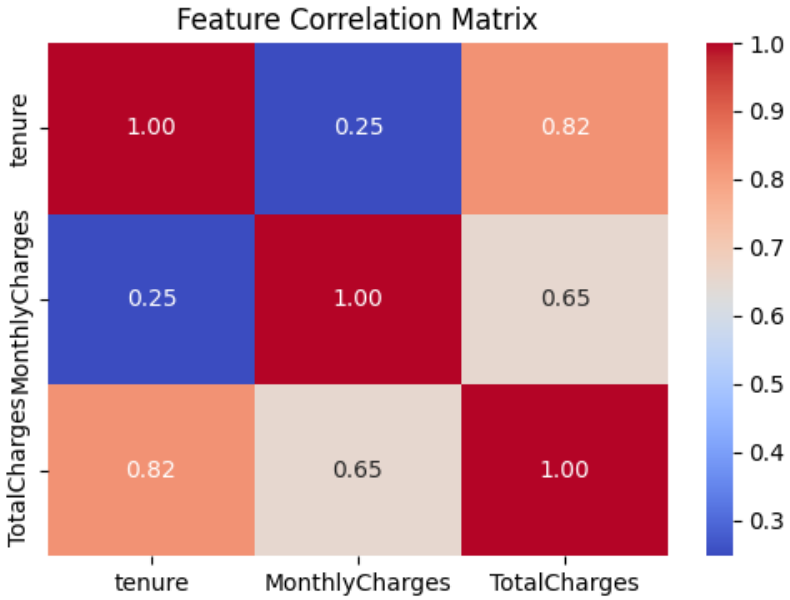

Figure 10: Correlation Matrix

10 Deployment

The trained model and preprocessing pipeline were integrated into a Gradio-based dashboard enabling real-time churn prediction and visualization.

11 Discussion

The project demonstrates a complete machine learning lifecycle including data processing, modeling, evaluation, and deployment. The modular design improves scalability and maintainability.

12 Conclusion

This project presents a production-ready telecom churn prediction system built using traditional machine learning techniques. The system integrates data processing, modeling, evaluation, and deployment to deliver an end-to-end predictive solution.

13 Future Work

- Retrieval-Augmented Generation integration using telecom knowledge base

- Agentic workflow development for prediction and recommendation orchestration
- LLM-based churn explanation module
- Vector database creation for customer interaction knowledge
- Multi-agent decision support system
- Hallucination mitigation strategies
- Conversational retention assistant
- Cloud-based scalable deployment

14 References

- Scikit-Learn Documentation
- Imbalanced-Learn Documentation
- Teleco Churn Dataset Documentation